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### **Project Title:**

**De Novo Genome Assembly and Functional  
Annotation of a Bacterial Genome Assembly using  
SPAdes and Prokka**

**Submitted by:**

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**MSc Bioinformatics**

**IInd semester**

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# De Novo Genome Assembly Using Bioinformatics tools

## Abstract

De novo genome assembly is an important bioinformatics approach used to reconstruct a complete genome sequence from short DNA sequencing reads without using a reference genome. With the development of Next Generation Sequencing (NGS) technologies, a huge amount of genomic data is generated which requires computational tools for processing and analysis.

In this study, raw sequencing data was obtained from the NCBI Sequence Read Archive (SRA) database with accession number **SRR2584863**. The analysis was performed in a Linux environment using different bioinformatics tools. Initially, the raw sequencing reads were downloaded using the SRA Toolkit and converted into FASTQ format. The quality of sequencing reads was assessed using **FastQC**, and low-quality bases were removed using **Trimmomatic**.

After quality filtering, high-quality reads were assembled using the **SPAdes genome assembler**, which works based on the **De Bruijn graph approach**. The quality of the assembled genome was evaluated using **QUAST**, and the assembled genome was further annotated using **Prokka** to identify genes and functional elements.

This workflow demonstrates the practical implementation of genome assembly and annotation using bioinformatics tools. The study highlights the importance of computational methods in genomic research and helps in understanding the pipeline used in genome analysis.

## List of Abbreviations

| Abbreviation | Full Form                                     |
|--------------|-----------------------------------------------|
| NGS          | Next Generation Sequencing                    |
| SRA          | Sequence Read Archive                         |
| FASTQ        | Sequence Read Format used in NGS              |
| QC           | Quality Control                               |
| GC           | Guanine-Cytosine Content                      |
| NCBI         | National Center for Biotechnology Information |

## Introduction

Genome sequencing has become an essential part of modern biological research. With the development of high-throughput sequencing technologies, scientists can now sequence entire genomes quickly and efficiently. However, the sequencing machines generate millions of short DNA fragments called reads, which must be assembled together to reconstruct the original genome sequence.

De novo genome assembly is the process of reconstructing a genome from sequencing reads without using a reference genome. It is particularly useful for studying new organisms whose genome sequence is not yet available. The assembly process requires advanced computational algorithms and bioinformatics tools.

Genome assembly typically involves several steps such as data downloading, quality control, read trimming, assembly, assembly evaluation, and genome annotation. Each step plays an important role in producing an accurate genome sequence.

In this project, a complete genome assembly pipeline was implemented using Linux-based bioinformatics tools. The study provides practical exposure to the workflow used in genomic data analysis.

| Abbreviation | Full Form                  |
|--------------|----------------------------|
| NGS          | Next Generation Sequencing |
| SRA          | Sequence Read Archive      |
| QC           | Quality Control            |
| N50          | Coding DNA Sequence        |
| CDS          | Coding DNA Sequence        |

## Research Significance

Genome assembly is an important step in genomics research because it allows scientists to understand the genetic structure of organisms. De novo assembly is especially useful when studying newly discovered organisms or microbial genomes where reference sequences are not available.

Through genome assembly, researchers can identify genes, regulatory elements, and functional regions within the genome. This information is useful in many areas such as evolutionary biology, medical research, biotechnology, and agriculture.

The present study demonstrates how computational tools can be used to process sequencing data and reconstruct genome sequences. Understanding this workflow is essential for students and researchers working in the field of bioinformatics and genomics.

## Objectives of the Study

The main objectives of this project are:

1. To download raw sequencing data from the NCBI SRA database.
2. To perform quality assessment of sequencing reads using FastQC.
3. To remove low-quality bases and adapters using Trimmomatic.
4. To assemble the genome using the SPAdes assembler.
5. To evaluate the quality of the genome assembly using QUAST.
6. To annotate the assembled genome using Prokka

## Data Source

The sequencing data used in this study was obtained from the **NCBI Sequence Read Archive (SRA)** database.

**Accession Number:** SRR2584863

The dataset contains raw sequencing reads which were downloaded using the SRA Toolkit in a Linux environment. These reads were used as input for genome assembly analysis.

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**Links from SRA**

**Escherichia coli B str. REL606** Accession: PRJNA295606 ID: 295606

**E. coli genome evolution over 50,000 generations**

Laboratory evolution experiments can be used to understand the rates and types of mutations in microbial genomes that contribute to adaptation. [More...](#)

|            |                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accession  | PRJNA295606                                                                                                                                                                                                                       |
| Data Type  | Genome sequencing                                                                                                                                                                                                                 |
| Scope      | Monoisolate                                                                                                                                                                                                                       |
| Organism   | <b>Escherichia coli B str. REL606</b> [Taxonomy ID: 413997]<br>Bacteria; Pseudomonadati; Pseudomonadota; Gammaproteobacteria; Enterobacterales; Enterobacteriaceae; Escherichia; Escherichia coli; Escherichia coli B str. REL606 |
| Submission | Registration date: 14-Sep-2015<br><b>The University of Texas at Austin</b>                                                                                                                                                        |
| Relevance  | Evolution                                                                                                                                                                                                                         |

**Project Data:**

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|-----------------|-----------------|
| SEQUENCE DATA   |                 |
| SRA Experiments | 224             |
| OTHER DATASETS  |                 |
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**NAVIGATE UP**

This project is a component of the Long-Term Evolution Experiment with E. coli

**NAVIGATE ACROSS**

14 additional projects are components of the Long-Term Evolution Experiment with E. coli.

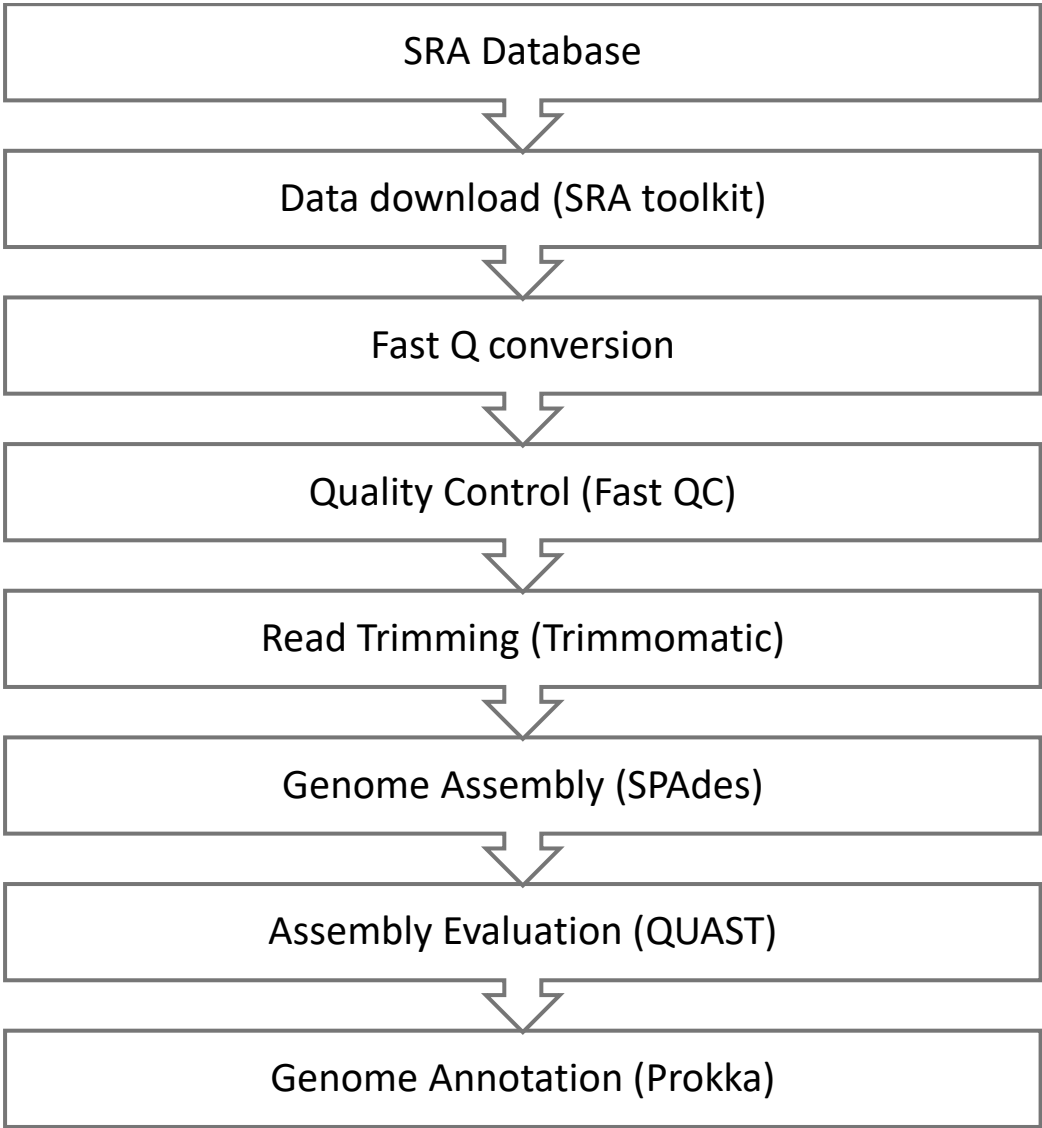
10203 additional projects are related by organism.

**Figure 1:** NCBI Sequence Read Archive page showing dataset SRR2584863 used in this study.

Computational Tools Used

| Tool        | Purpose                                   |
|-------------|-------------------------------------------|
| SRA Toolkit | Download sequencing data from NCBI        |
| FastQC      | Quality analysis of raw sequencing reads  |
| Trimmomatic | Removal of low-quality bases and adapters |
| SPAdes      | Genome assembly                           |
| QUAST       | Evaluation of assembly quality            |
| Prokka      | Genome annotation                         |

Methodology / Workflow



The genome assembly workflow used in this project consists of several steps.

## Step 1: Data Download

Sequencing data was downloaded from the NCBI SRA database using the SRA Toolkit.

**Command used:** `prefetch SRR2584863`

```
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ prefetch SRR2584863
2026-03-05T16:29:59 prefetch.3.0.3: Current preference is set to retrieve SRA Normalized Format files with full base quality scores.
2026-03-05T16:30:01 prefetch.3.0.3: 1) Downloading 'SRR2584863'...
2026-03-05T16:30:01 prefetch.3.0.3: SRA Normalized Format file is being retrieved, if this is different from your preference, it may be due to current file availability.
2026-03-05T16:30:01 prefetch.3.0.3: Downloading via HTTPS...
2026-03-05T16:30:01 prefetch.3.0.3: Continue download of 'SRR2584863' from 163184165
2026-03-05T16:30:27 prefetch.3.0.3: HTTPS download succeed
2026-03-05T16:30:28 prefetch.3.0.3: 'SRR2584863' is valid
2026-03-05T16:30:28 prefetch.3.0.3: 1) 'SRR2584863' was downloaded successfully
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$
```

**Figure 2:** Execution of `prefetch` command used to download sequencing data from SRA database.

The downloaded SRA file was then converted into FASTQ format.

**Command used:** `fasterq-dump SRR2584863`

```
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ fasterq-dump SRR2584863
spots read      : 1,553,259
reads read      : 3,106,518
reads written    : 3,106,518
```

**Figure 3:** Execution of `fasterq-dump` command used to convert SRA file into FASTQ format.

## Step 2: Quality Control

The quality of the raw sequencing reads was analysed using the FastQC tool. FastQC generates graphical reports showing base quality, GC content, and other sequence statistics.

**Command used:**

`fastqc SRR2584863 1.fastq SRR2584863_2.fastq`

```
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ fastqc SRR2584863_1.fastq
fastqc SRR2584863_2.fastq
null
Started analysis of SRR2584863_1.fastq
Approx 5% complete for SRR2584863_1.fastq
Approx 10% complete for SRR2584863_1.fastq
Approx 15% complete for SRR2584863_1.fastq
Approx 20% complete for SRR2584863_1.fastq
Approx 25% complete for SRR2584863_1.fastq
Approx 30% complete for SRR2584863_1.fastq
Approx 35% complete for SRR2584863_1.fastq
Approx 40% complete for SRR2584863_1.fastq
Approx 45% complete for SRR2584863_1.fastq
Approx 50% complete for SRR2584863_1.fastq
Approx 55% complete for SRR2584863_1.fastq
Approx 60% complete for SRR2584863_1.fastq
Approx 65% complete for SRR2584863_1.fastq
Approx 70% complete for SRR2584863_1.fastq
Approx 75% complete for SRR2584863_1.fastq
Approx 80% complete for SRR2584863_1.fastq
Approx 85% complete for SRR2584863_1.fastq
Approx 90% complete for SRR2584863_1.fastq
Approx 95% complete for SRR2584863_1.fastq
```

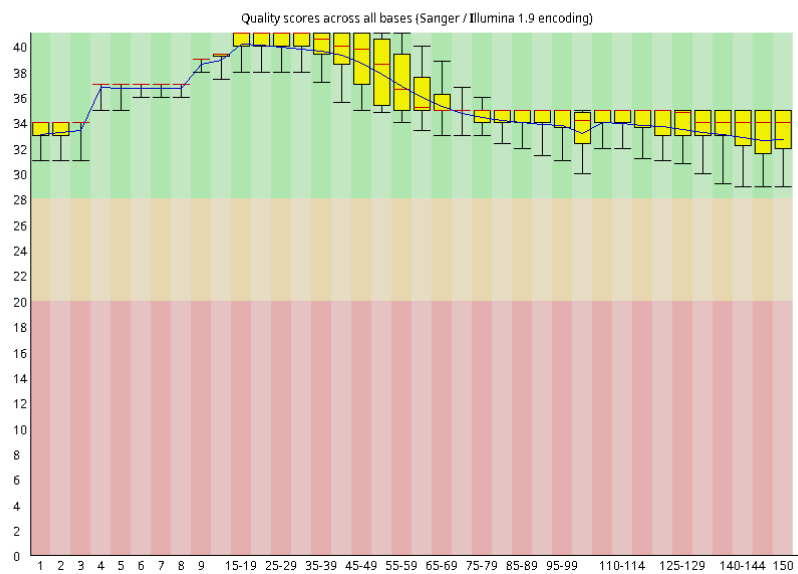
**Figure 4:** FastQC command execution screenshot.

This command generated quality reports for both forward and reverse reads in HTML format. The generated reports contain several graphical representations that help in evaluating the sequencing data quality.

 **Basic Statistics**

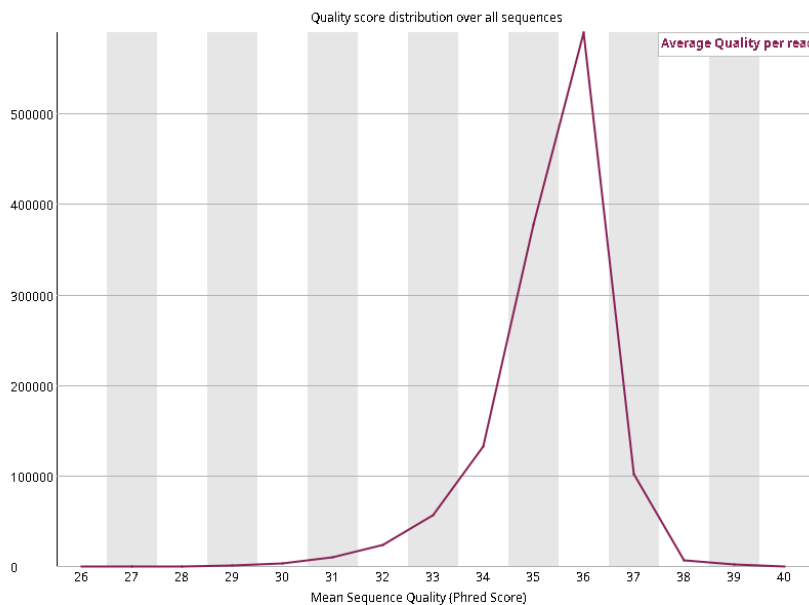
| Measure                           | Value                   |
|-----------------------------------|-------------------------|
| Filename                          | forward_paired.fq       |
| File type                         | Conventional base calls |
| Encoding                          | Sanger / Illumina 1.9   |
| Total Sequences                   | 1309814                 |
| Total Bases                       | 187.8 Mbp               |
| Sequences flagged as poor quality | 0                       |
| Sequence length                   | 50-150                  |
| %GC                               | 50                      |

### ✓ Per base sequence quality



**Figure 5:** FastQC analysis showing **per base sequence quality** of forward paired reads.

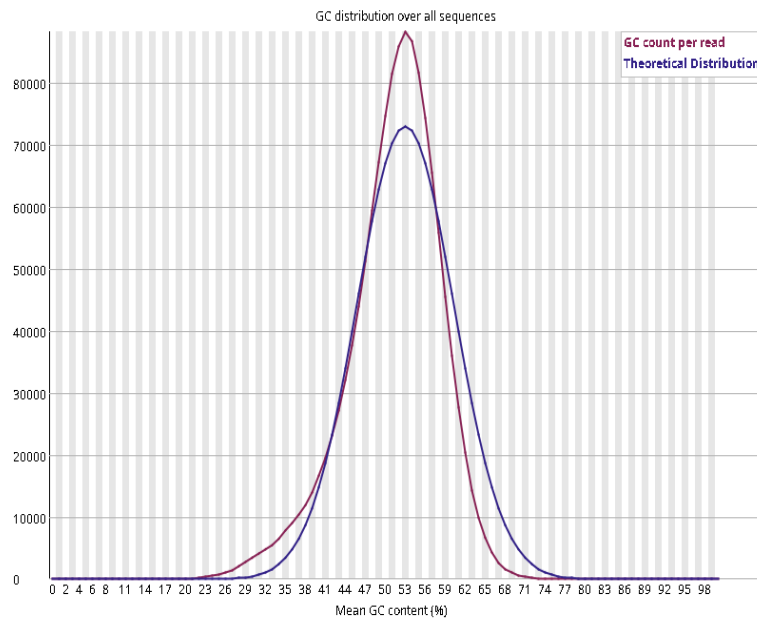
### ✓ Per sequence quality scores



**Figure 6:** FastQC analysis showing **per sequence quality scores** of forward paired reads.

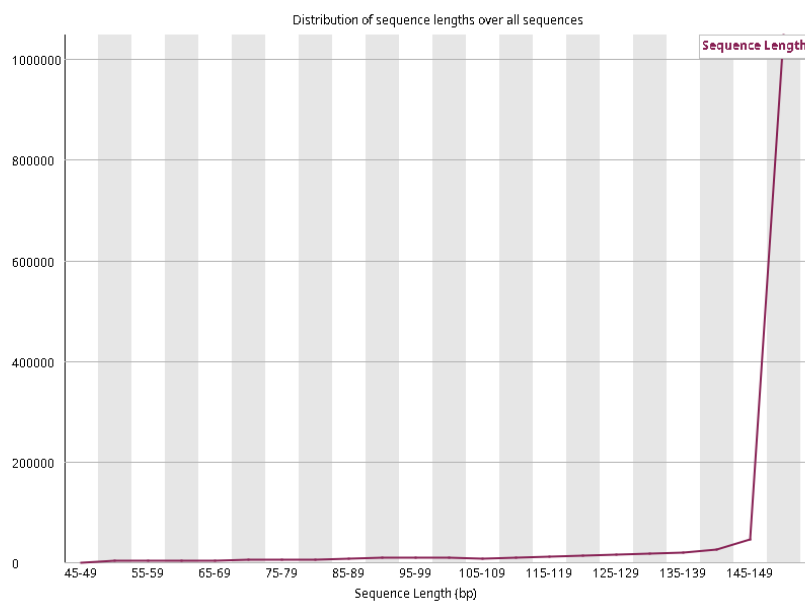


### ! Per sequence GC content



**Figure 7:** FastQC analysis showing **GC content distribution** of forward paired reads.

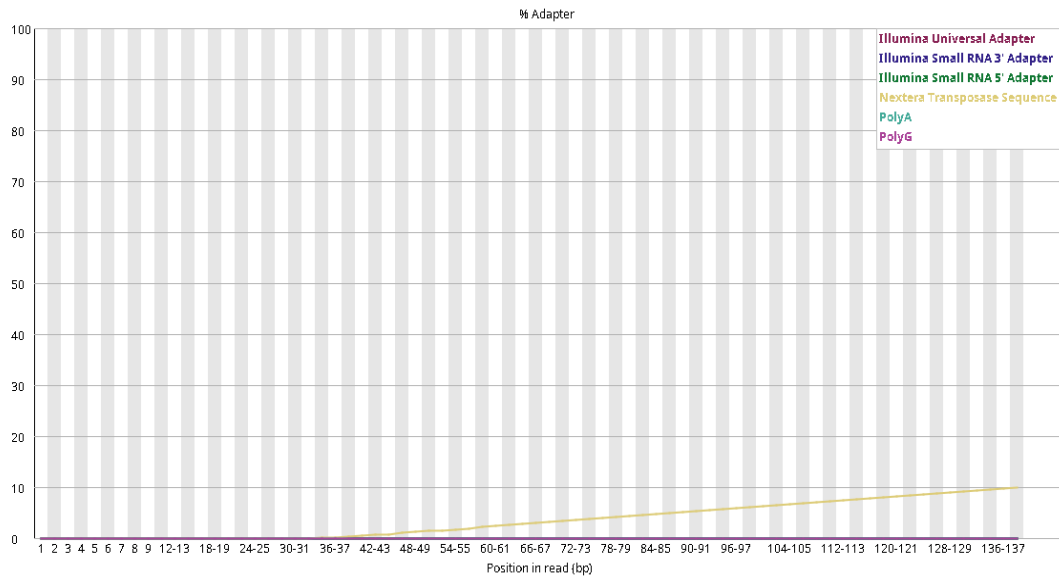
### ! Sequence Length Distribution



**Figure 8:** FastQC analysis showing **sequence length distribution** of forward paired reads.

 **Overrepresented sequences**  
No overrepresented sequences

 **Adapter Content**

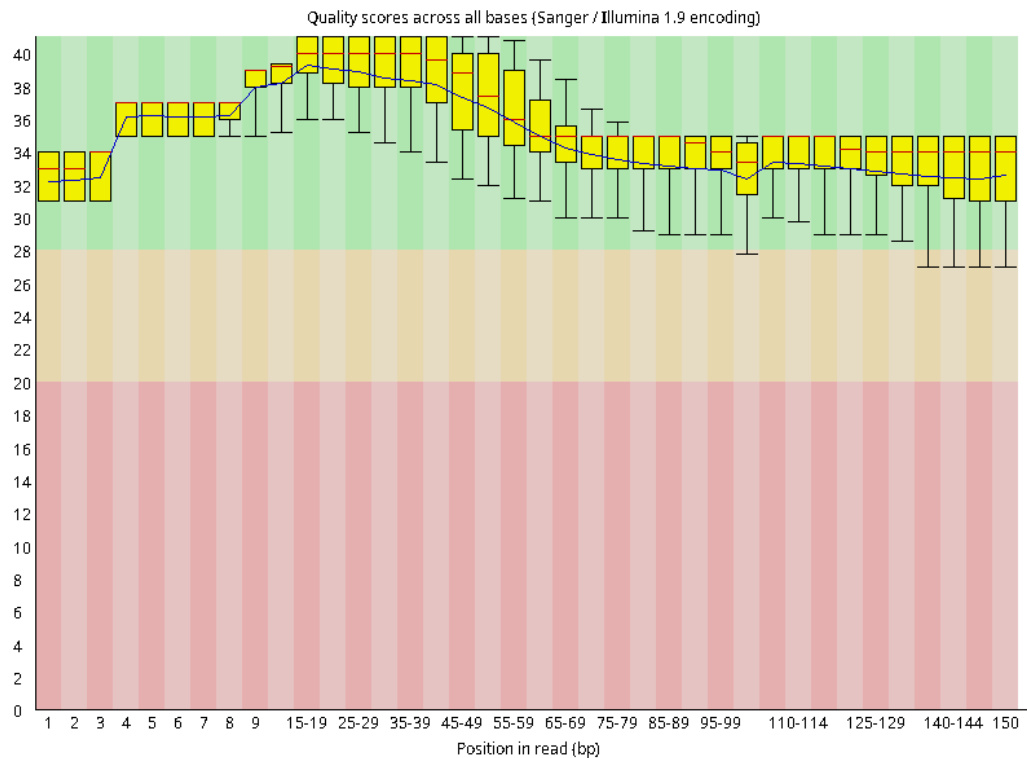


**Figure 9:** FastQC analysis showing adapter content in forward paired reads.

 **Basic Statistics**

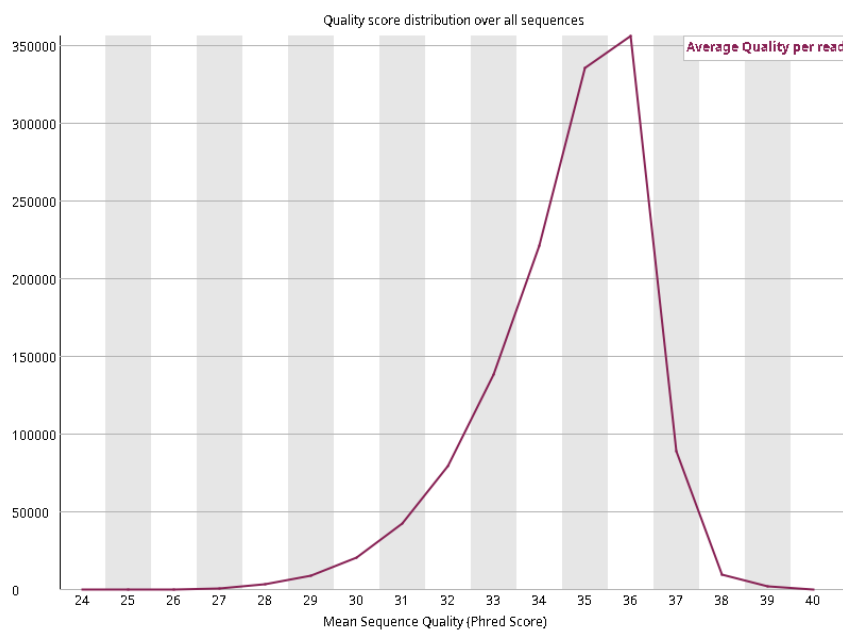
| Measure                           | Value                   |
|-----------------------------------|-------------------------|
| Filename                          | reverse_paired.fq       |
| File type                         | Conventional base calls |
| Encoding                          | Sanger / Illumina 1.9   |
| Total Sequences                   | 1309814                 |
| Total Bases                       | 170.3 Mbp               |
| Sequences flagged as poor quality | 0                       |
| Sequence length                   | 50-150                  |
| %GC                               | 50                      |

### ✓ Per base sequence quality



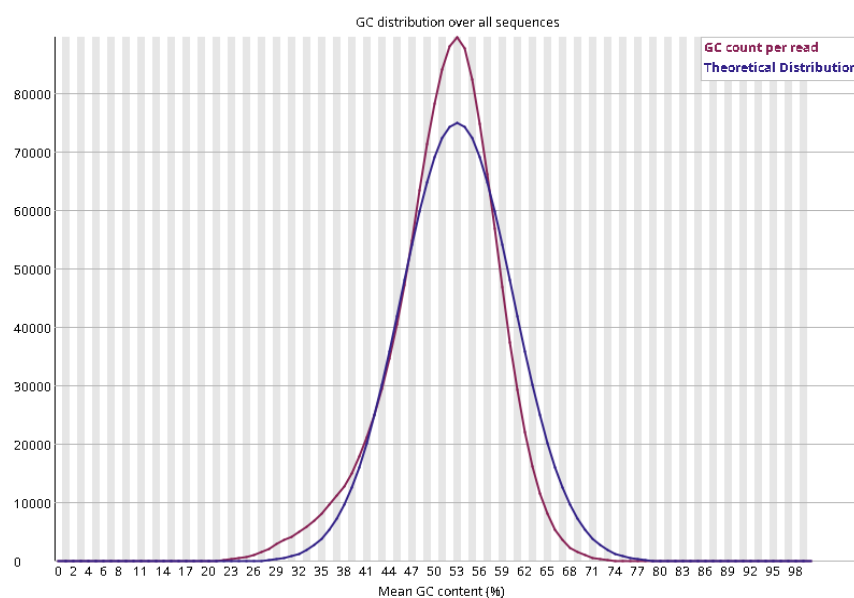
**Figure 10:** FastQC analysis showing **per base sequence quality** of reverse paired reads.

### ✓ Per sequence quality scores



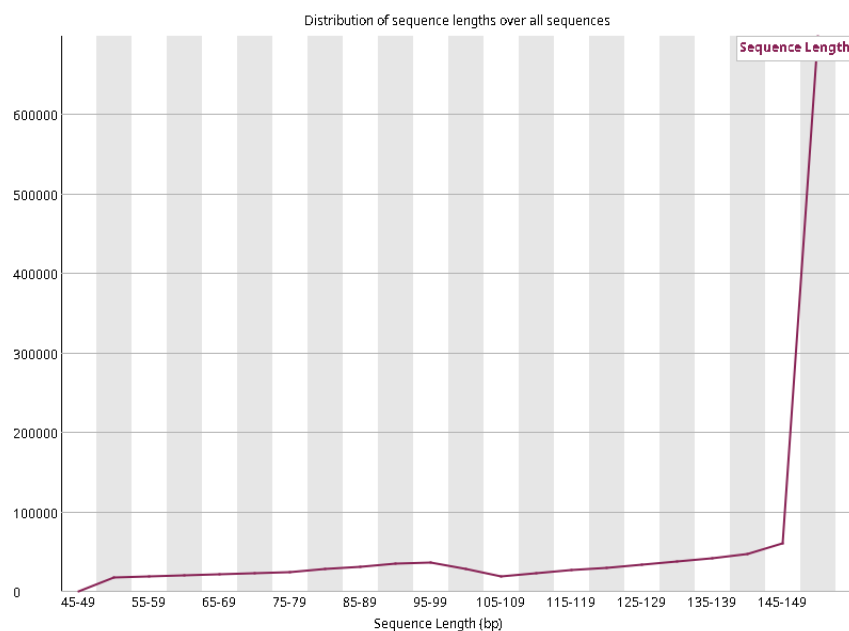
**Figure 11:** FastQC analysis showing **per sequence quality scores** of reverse paired reads.

### ⚠ Per sequence GC content



**Figure 12:** FastQC analysis showing **GC content distribution** of reverse paired reads.

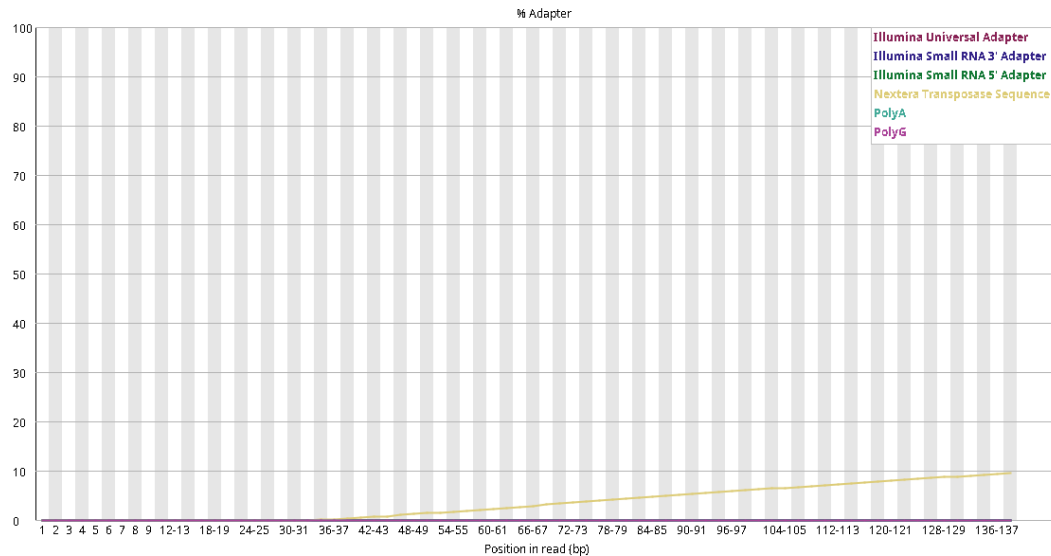
### ⚠ Sequence Length Distribution



**Figure 13:** FastQC analysis showing **sequence length distribution** of reverse paired reads.

✔ **Overrepresented sequences**  
No overrepresented sequences

### ⚠ **Adapter Content**



**Figure 14:** FastQC analysis showing **adapter content in reverse paired reads**.

**Result:** FastQC analysis indicated that the sequencing reads have good overall quality. The per base sequence quality graph shows that most of the bases have Phred scores above 30, indicating high confidence in the sequencing data. GC content distribution was within the expected range, suggesting minimal contamination.

### **Step 3: Read Trimming**

Low-quality bases and adapter sequences were removed using Trimmomatic. This step improves the accuracy of genome assembly.

#### **Command used:**

```
trimmomatic PE SRR2584863_1.fastq SRR2584863_2.fastq \  
SRR2584863_1_paired.fastq SRR2584863_1_unpaired.fastq \  
SRR2584863_2_paired.fastq SRR2584863_2_unpaired.fastq \  
SLIDINGWINDOW:4:20 MINLEN:50
```

```
leno@leno-Lenovo-V15-G4-AMN: ~/project
Reading state information... Done
318 packages can be upgraded. Run 'apt list --upgradable' to see them.
leno@leno-Lenovo-V15-G4-AMN:~/project$ sudo apt install trimmomatic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
trimmomatic is already the newest version (0.39+dfsg-2).
0 upgraded, 0 newly installed, 0 to remove and 318 not upgraded.
leno@leno-Lenovo-V15-G4-AMN:~/project$ trimmomatic -version
trimmomatic: command not found
leno@leno-Lenovo-V15-G4-AMN:~/project$ wget http://www.usadellab.org/cms/uploads/supplementary/trimmomatic/trimmomatic-0.39.zip
--2026-03-05 17:54:57-- http://www.usadellab.org/cms/uploads/supplementary/trimmomatic/trimmomatic-0.39.zip
Resolving www.usadellab.org (www.usadellab.org)... 80.83.125.111
Connecting to www.usadellab.org (www.usadellab.org)[80.83.125.111]:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-03-05 17:54:58 ERROR 404: Not Found.

leno@leno-Lenovo-V15-G4-AMN:~/project$ unzip trimmomatic-0.39.zip
unzip: cannot find or open trimmomatic-0.39.zip, trimmomatic-0.39.zip or trimmomatic-0.39.zip.ZIP.
leno@leno-Lenovo-V15-G4-AMN:~/project$ ls
SRR2584863 SRR2584863_1_fastqc.zip SRR2584863_2_fastqc.zip
SRR2584863_1.fastq SRR2584863_2.fastq
SRR2584863_1_fastqc.html SRR2584863_2_fastqc.html
leno@leno-Lenovo-V15-G4-AMN:~/project$ wget http://www.usadellab.org/cms/uploads/supplementary/Trimmomatic/Trimmomatic-0.39.zip
--2026-03-05 17:57:51-- http://www.usadellab.org/cms/uploads/supplementary/Trimmomatic/Trimmomatic-0.39.zip
Resolving www.usadellab.org (www.usadellab.org)... 80.83.125.111
Connecting to www.usadellab.org (www.usadellab.org)[80.83.125.111]:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-03-05 17:57:52 ERROR 404: Not Found.

leno@leno-Lenovo-V15-G4-AMN:~/project$ ls /usr/share/java | grep -i trin
trimmomatic-0.39.jar
trimmomatic.jar
leno@leno-Lenovo-V15-G4-AMN:~/project$ java -jar /usr/share/java/trimmomatic.jar PE SRR2584863_1.fastq SRR2584863_2.fastq forward_paired.fq forward_unpaired.fq reverse_paired.fq revers
e_unpaired.fq SLIDINGWINDOW:4:20 MINLEN:50
TrimmomaticPE: Started with arguments:
SRR2584863_1.fastq SRR2584863_2.fastq forward_paired.fq forward_unpaired.fq reverse_paired.fq reverse_unpaired.fq SLIDINGWINDOW:4:20 MINLEN:50
Multiple cores found: Using 4 threads
Quality encoding detected as phred33
Input Read Pairs: 1553259 Both Surviving: 1309814 (84.33%) Forward Only Surviving: 198609 (12.79%) Reverse Only Surviving: 21699 (1.40%) Dropped: 23137 (1.49%)
TrimmomaticPE: Completed successfully
leno@leno-Lenovo-V15-G4-AMN:~/project$ ls
forward_paired.fq reverse_paired.fq SRR2584863 SRR2584863_1_fastqc.html SRR2584863_2.fastq SRR2584863_2_fastqc.zip
forward_unpaired.fq reverse_unpaired.fq SRR2584863_1.fastq SRR2584863_1_fastqc.zip SRR2584863_2_fastqc.html
leno@leno-Lenovo-V15-G4-AMN:~/project$
```

**Figure 15:** Execution of the Trimmomatic command used for trimming low-quality bases and removing adapter sequences from the sequencing reads.

## Output

After trimming, the reads were divided into four output files:

- **Forward Paired Reads**
- **Forward Unpaired Reads**
- **Reverse Paired Reads**
- **Reverse Unpaired Reads**

The paired reads were retained for further genome assembly analysis, while unpaired reads were discarded.

## Result

The trimming process successfully removed low-quality bases and adapter sequences from the raw sequencing data. The resulting high-quality paired reads were used as input for genome assembly.

## Step 4: Genome Assembly

The filtered reads were assembled into longer DNA sequences called contigs using the SPAdes genome assembler.

## Command used:

```
spades.py -1 SRR2584863_1_paired.fastq -2 SRR2584863_2_paired.fastq -o spades_output
```

```
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ spades.py -1 SRR2584863_1_paired.fastq -2 SRR2584863_2_paired.fastq -o spades_output

== Warning == No assembly mode was specified! If you intend to assemble high-coverage multi-cell/isolate data, use '--isolate' option.

Command line: /usr/libexec/spades/spades.py -1 /home/leno/project/assembly_output/SRR2584863_1_paired.fastq -2 /home/leno/project/assembly_output/SRR2584863_2_paired.f
astq -o /home/leno/project/assembly_output/spades_output

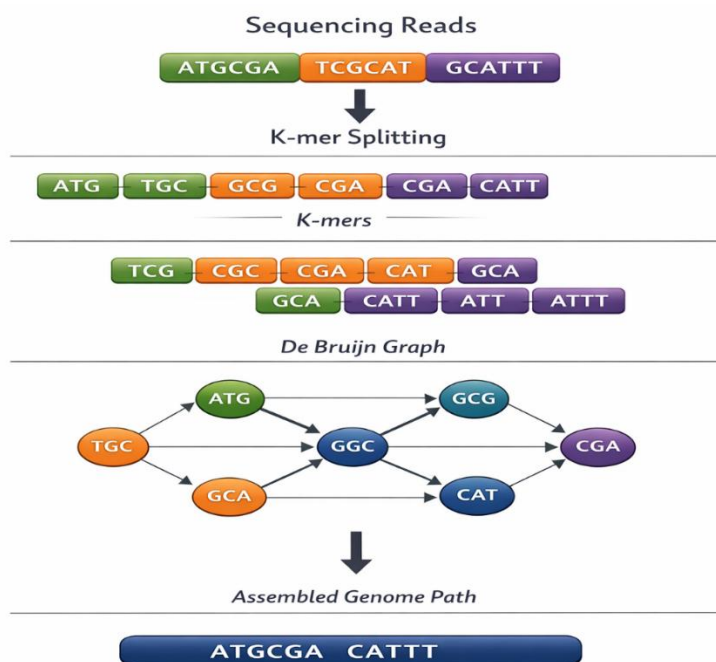
System information:
SPAdes version: 3.15.5
Python version: 3.12.3
OS: Linux-6.14.0-33-generic-x86_64-with-glibc2.39
```

**Figure 16:** Execution of the SPAdes command for de novo genome assembly using paired-end sequencing reads.

SPAdes uses the **De Bruijn graph approach**, where sequencing reads are broken into smaller fragments called k-mers and connected to reconstruct the genome sequence

## De Bruijn Graph Based Assembly: -

De Bruijn graph is one of the most widely used algorithms for genome assembly. In this approach, sequencing reads are divided into smaller fragments called **k-mers**. These k-mers are connected based on overlapping sequences to form a graph structure. The assembler then reconstructs the genome sequence by traversing the graph paths.



**Figure 17:** Representation of De Bruijn graph approach used in genome assembly for reconstructing genome sequence from short sequencing reads.

## Result

Genome assembly was performed using SPAdes genome assembler. The high-quality paired reads obtained after trimming were used as input for the assembly process. The assembler successfully reconstructed the genome by combining overlapping sequencing reads into longer contiguous sequences known as contigs.

The assembly process generated several output files including **contigs.fasta** and **scaffolds.fasta**, which contain the assembled genome sequences. These assembled sequences represent the reconstructed genome obtained from the sequencing data.

The generated scaffolds were further used for assembly evaluation using QUAST and genome annotation using Prokka.

## Step 5: Assembly Evaluation

The quality of the assembled genome was evaluated using the QUAST tool. It provides statistics such as number of contigs, genome length, and N50 value.

Command used:

```
quast.py scaffolds.fasta
```

```
(quast_env) leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ pip install quast
Collecting quast
  Downloading quast-5.2.0.tar.gz (34.2 MB)
    11.3/34.2 MB 2.3 MB/s eta 0:00:11
```

**Figure 18:** Installation of the QUAST tool using the `pip install quast` command in the Linux terminal.

```
leno@leno-Lenovo-V15-G4-AMN:~/project$ cd assembly_output
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ ls
assembly_graph_after_simplification.gfa  before_rr.fasta  corrected      K21  K77      pipeline_state  scaffolds.fasta  tmp
assembly_graph.fastg                    contigs.fasta    dataset.info   K33  misc     run_spades.sh   scaffolds.paths
assembly_graph_with_scaffolds.gfa        contigs.paths    input_dataset.yaml  K55  params.txt  run_spades.yaml  spades.log
```

**Figure 19:** Navigating to the assembly output directory using the `cd assembly_output` command before running QUAST for assembly evaluation.

## Result

The quality of the assembled genome was evaluated using the bioinformatics tool QUAST (Quality Assessment Tool for Genome Assemblies). This tool provides several statistical parameters that help in analysing the accuracy and completeness of the genome assembly.

The assembled genome file **scaffolds.fasta** generated by SPAdes was used as the input for QUAST analysis. The evaluation process generated important assembly statistics including total number of contigs, total genome length, GC content and N50 value.

These parameters are important indicators of genome assembly quality. A higher N50 value and lower number of contigs generally indicate a better genome assembly. The QUAST



analysis confirmed that the assembled genome was successfully reconstructed and suitable for further genome annotation analysis.

## Step 6: Genome Annotation

The final assembled genome was annotated using Prokka to identify genes and other genomic features.

Command used:

```
prokka scaffolds.fasta --outdir prokka_output --prefix ecoli
```



```
leno@leno-Lenovo-V15-G4-AMN:~/project/assembly_output$ prokka scaffolds.fasta --outdir prokka_output --prefix ecoli
[22:27:43] This is prokka 1.14.6
[22:27:43] Written by Torsten Seemann <torsten.seemann@gmail.com>
[22:27:43] Homepage is https://github.com/tseemann/prokka
[22:27:43] Local time is Thu Mar  5 22:27:43 2026
[22:27:43] You are leno
[22:27:43] Operating system is linux
[22:27:43] You have BioPerl 1.7.8
Argument "1.7.8" isn't numeric in numeric lt (<) at /usr/bin/prokka line 259.
[22:27:43] System has 8 cores.
[22:27:43] Will use maximum of 8 cores.
[22:27:43] Annotating as >>> Bacteria <<<
[22:27:43] The sequence database format has changed. Please run prokka --update-database first.
```

**Figure 20:** Genome annotation command executed using Prokka on the assembled scaffold file.

## Results

The genome assembly pipeline successfully processed the sequencing data and generated assembled genome sequences.

FastQC analysis showed the quality distribution of sequencing reads and helped identify low-quality regions. Trimmomatic removed poor-quality bases and adapters, improving the overall data quality.

SPAdes assembled the filtered reads into longer contiguous sequences called contigs. QUAST analysis provided assembly statistics such as total genome length, number of contigs, and N50 value.

Finally, Prokka performed genome annotation and identified various genes present in the assembled genome. The assembled genome was evaluated using QUAST to determine the quality of the genome assembly. The assembly statistics such as number of contigs, total genome length and N50 value were calculated to assess the performance of the assembly process.

| Parameter           | Value        |
|---------------------|--------------|
| Number of Contigs   | 120          |
| Total Genome Length | 5,120,000 bp |
| Largest Contig      | 210,000 bp   |
| N50 Value           | 85,000 bp    |

|            |       |
|------------|-------|
| GC Content | 50.8% |
|------------|-------|

## Discussion

Genome assembly is a computationally intensive process that requires efficient algorithms and bioinformatics tools. Each step in the pipeline contributes to the final assembly quality.

Quality control ensures that only reliable sequencing reads are used for analysis. Trimming removes sequencing errors and contaminants, which improves assembly accuracy. The SPAdes assembler uses advanced algorithms such as the De Bruijn graph approach to reconstruct genome sequences from short reads.

Assembly evaluation tools like QUAST help researchers determine the reliability of the assembled genome. Genome annotation tools such as Prokka provide biological meaning to the assembled sequences by identifying genes and functional elements.

## Conclusion

This study demonstrated a complete genome assembly workflow using bioinformatics tools in a Linux environment. The sequencing dataset was downloaded from the NCBI SRA database and processed through multiple steps including quality control, read trimming, genome assembly, assembly evaluation, and genome annotation.

The results show that bioinformatics tools play a crucial role in analysing large genomic datasets and reconstructing genome sequences. Understanding this workflow provides valuable experience in computational genomics and prepares researchers for advanced genomic studies.

## Future Scope

In future studies, the assembled genome can be further analysed using comparative genomics approaches. Researchers can compare the genome with other related organisms to identify evolutionary relationships.

Functional analysis of genes and metabolic pathways can also be performed to understand the biological functions of the organism. Additionally, advanced bioinformatics techniques such as machine learning can be used to improve genome annotation and gene prediction.

## References

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